Dynamic Response 4

Professor: Juan David Bastidas Rodríguez

Reference book for the presentation: **Feedback Control of Dynamic Systems, 8th Edition, Franklin • Powell • Emami-Naeini, Pearson, 2019, ISBN: 978-0-13-468571-7**

Content

1. Effects of Zeros and Additional Poles
2. Stability

1. Effects of Zeros and Additional Poles

- Zeros modify the coefficients of the exponential terms of the impulse response
- A **zero near a pole** reduces the **amount** of that term in the total response
- Calculating the coefficients for a partial-fraction expansion:
  - If F(s) has a zero near the pole at s = p1, the value of F(s) will be small (s is near the zero)

$$C_1 = (s - p_1)F(s)|_{s=p_1}$$

1. Effects of Zeros and Additional Poles

Example:

$$H_1(s) = \frac{2}{(s+1)(s+2)} = \frac{2}{s+1} - \frac{2}{s+2}$$

$$H_2(s) = \frac{2(s+1.1)}{1.1(s+1)(s+2)} = \frac{2}{1.1}\left(\frac{0.1}{s+1} + \frac{0.9}{s+2}\right) = \frac{0.18}{s+1} + \frac{1.64}{s+2}.$$

This dramatic reduction is brought about by the zero at s = −1.1 in H2(s), which almost cancels the pole at s = −1

1. Effects of Zeros and Additional Poles

- 2nd order system with one zero
- The zero is located at s = −αζωn = −ασ
- If α is large, the zero will be far removed from the poles (little effect)
- If α ≈ 1, the value of the zero will be close to the real part of the poles (substantial influence)

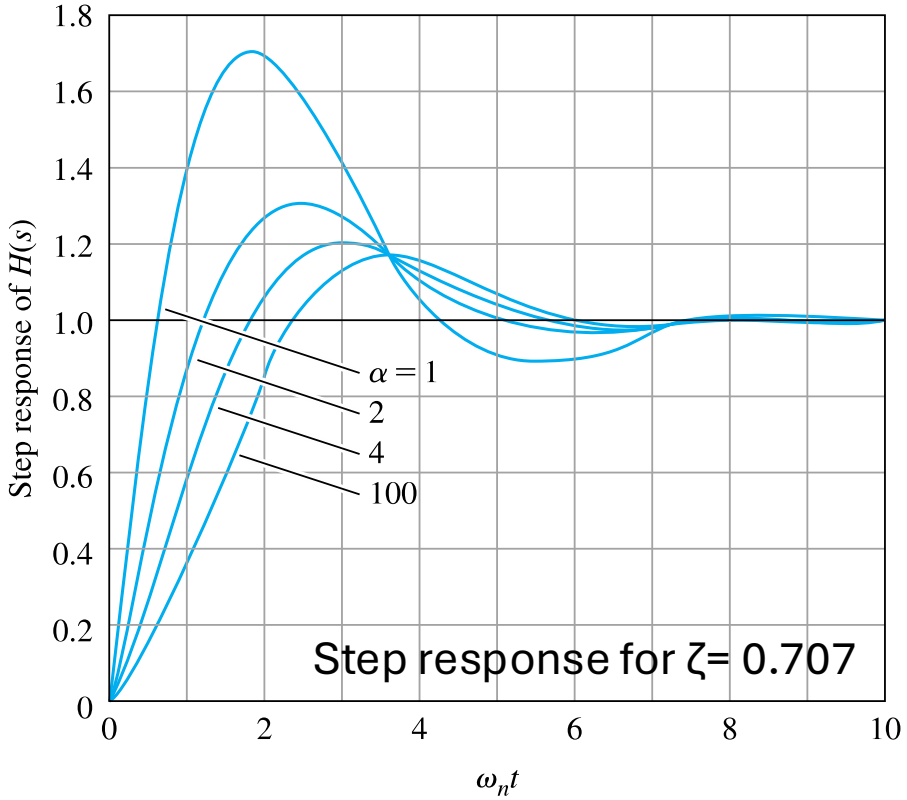
$$H(s) = \frac{(s/\alpha\zeta\omega_n) + 1}{(s/\omega_n)^2 + 2\zeta(s/\omega_n) + 1}$$

1. Effects of Zeros and Additional Poles

2nd order system with one zero

$$H(s) = \frac{(s/\alpha\zeta\omega_n) + 1}{(s/\omega_n)^2 + 2\zeta(s/\omega_n) + 1}$$

- Increase the overshoot Mp
- Reduce rise time tr
- Very little influence on the settling time ts

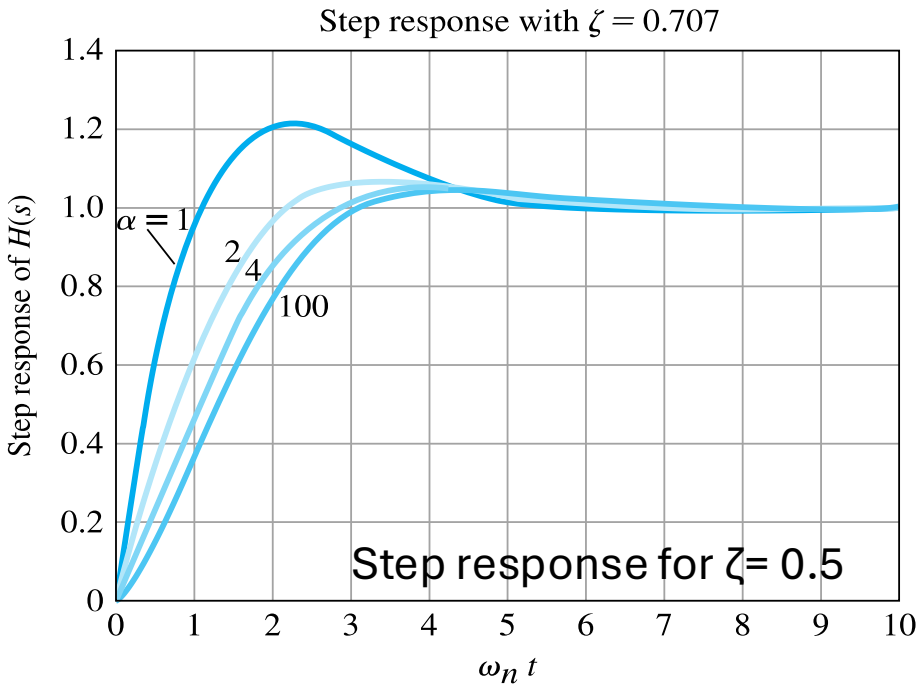


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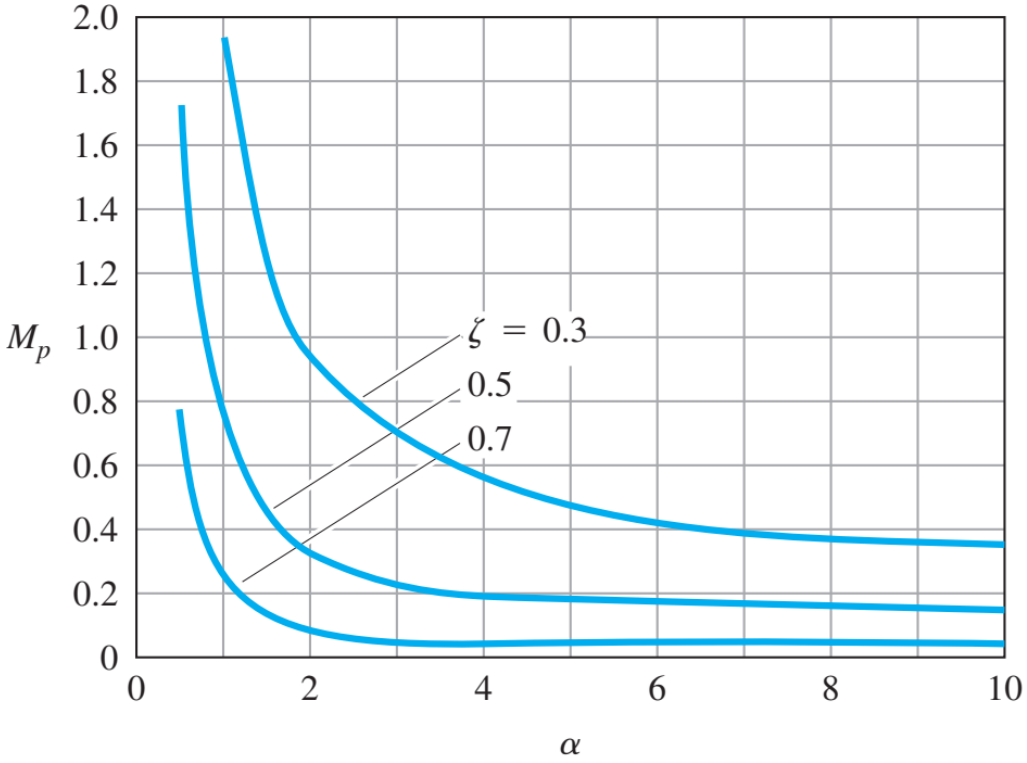


1. Effects of Zeros and Additional Poles

2nd order system with one zero

$$H(s) = \frac{(s/\alpha\zeta\omega_n) + 1}{(s/\omega_n)^2 + 2\zeta(s/\omega_n) + 1}$$

- **Very little effect** on Mp if α > 3
- **Increasing effect** but as α < 3
- **High effect** for α = 1 or less



1. Effects of Zeros and Additional Poles

- 2nd order system with one zero

$$H(s) = \frac{(s/\alpha\zeta\omega_n) + 1}{(s/\omega_n)^2 + 2\zeta(s/\omega_n) + 1}$$

- Replacing s/wn by s

$$H(s) = \frac{s/\alpha\zeta + 1}{s^2 + 2\zeta s + 1}.$$

- Rewriting the equation
  - H0(s) is the original term (no zero),
  - Hd(s) (introduced by the zero) is H0(s)(1/αζ)s (i.e. derivative)

$$H(s) = \frac{1}{s^2 + 2\zeta s + 1} + \frac{1}{\alpha\zeta}\frac{s}{s^2 + 2\zeta s + 1}$$

- In time domain

$$y(t) = y_0(t) + y_d(t) = y_0(t) + \frac{1}{\alpha\zeta}\dot{y}_0(t)$$

1. Effects of Zeros and Additional Poles

2nd order system with one zero

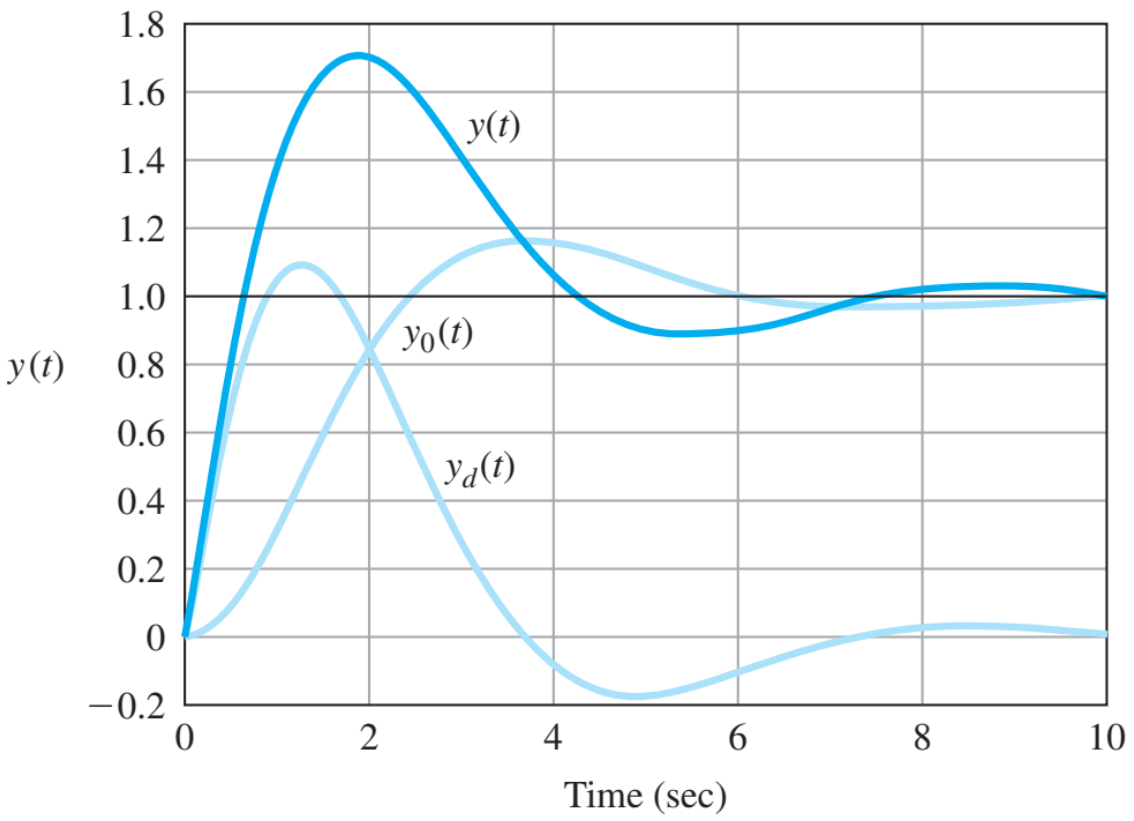
- **If the zero is in the LHP**

$$H(s) = \frac{(s/\alpha\zeta\omega_n) + 1}{(s/\omega_n)^2 + 2\zeta(s/\omega_n) + 1}$$

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1. Effects of Zeros and Additional Poles

2nd order system with one zero

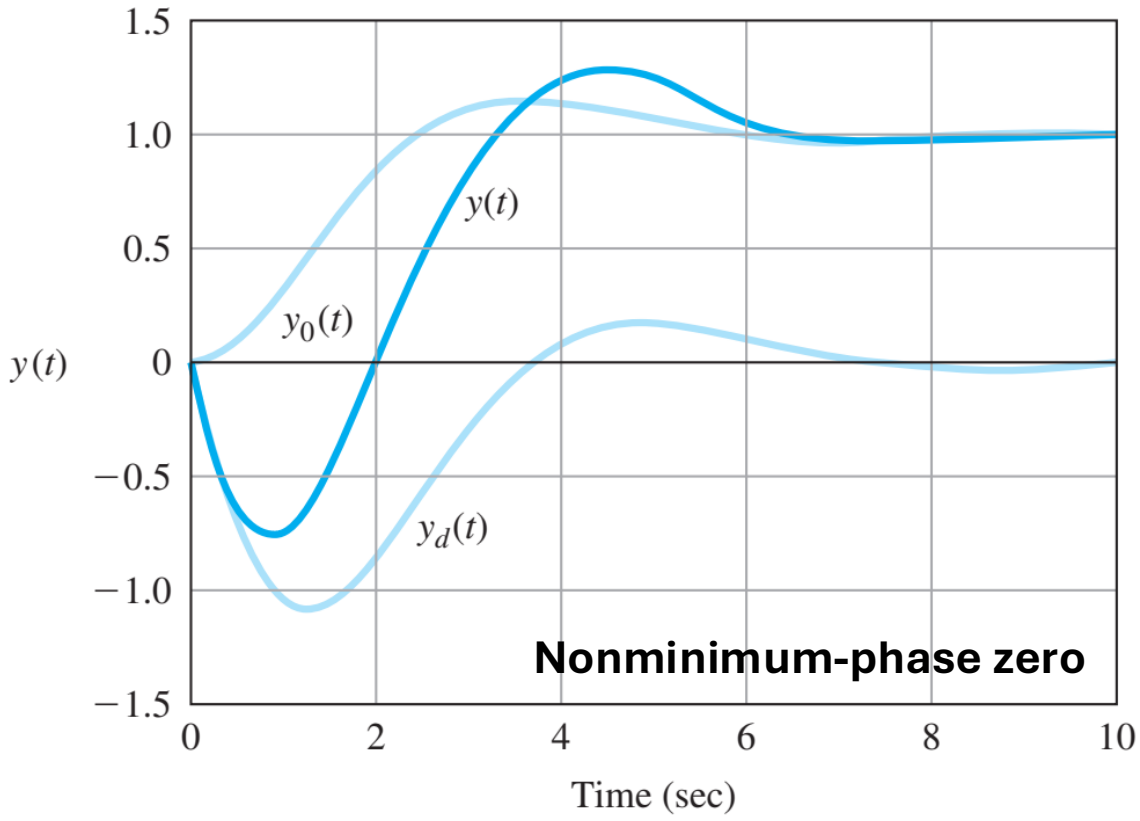
- **If the zero is in the RHP**

$$H(s) = \frac{(s/\alpha\zeta\omega_n) + 1}{(s/\omega_n)^2 + 2\zeta(s/\omega_n) + 1}$$

$$H(s) = \frac{s/\alpha\zeta + 1}{s^2 + 2\zeta s + 1}.$$

$$H(s) = \frac{1}{s^2 + 2\zeta s + 1} + \frac{1}{\alpha\zeta}\frac{s}{s^2 + 2\zeta s + 1}$$

$$y(t) = y_0(t) + y_d(t) = y_0(t) + \frac{1}{\alpha\zeta}\dot{y}_0(t)$$



1. Effects of Zeros and Additional Poles

Example: Effect of the Proximity of the Zero to the Pole Locations on the Transient Response

- 2nd order system with a finite zero and unity DC gain
- Determine the effect of the zero location (s = −z) on the unit-step response when z = {1, 2, 3, 4, 5, 6}

$$H(s) = \frac{24}{z}\frac{(s+z)}{(s+4)(s+6)}$$

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$$H(s) = \frac{24}{z}\frac{(s+z)}{(s+4)(s+6)}$$

$$H_1(s) = H(s)\frac{1}{s} = \frac{24}{z}\frac{(s+z)}{s(s+4)(s+6)}$$

$$= \frac{24}{z}\frac{s}{s(s+4)(s+6)} + \frac{24}{s(s+4)(s+6)}$$

Applying the inverse Laplace Trasform

$$y(t) = y_1(t) + y_2(t),$$

1. Effects of Zeros and Additional Poles

Example: Effect of the Proximity of the Zero to the Pole Locations on the Transient Response

- If z = 4 or z = 6 => pole–zero cancellations
- The response as a 1st order

$$H_1(s) = \frac{24}{z}\frac{s}{s(s+4)(s+6)} + \frac{24}{s(s+4)(s+6)}$$

Applying the inverse Laplace Trasform

$$y(t) = y_1(t) + y_2(t),$$

$$y_1(t) = \frac{12}{z}e^{-4t} - \frac{12}{z}e^{-6t},$$

$$y_2(t) = z\int_0^t y_1(\tau)d\tau = -3e^{-4t} + 2e^{-6t} + 1,$$

$$y(t) = 1 + \left(\frac{12}{z} - 3\right)e^{-4t} + \left(2 - \frac{12}{z}\right)e^{-6t}$$

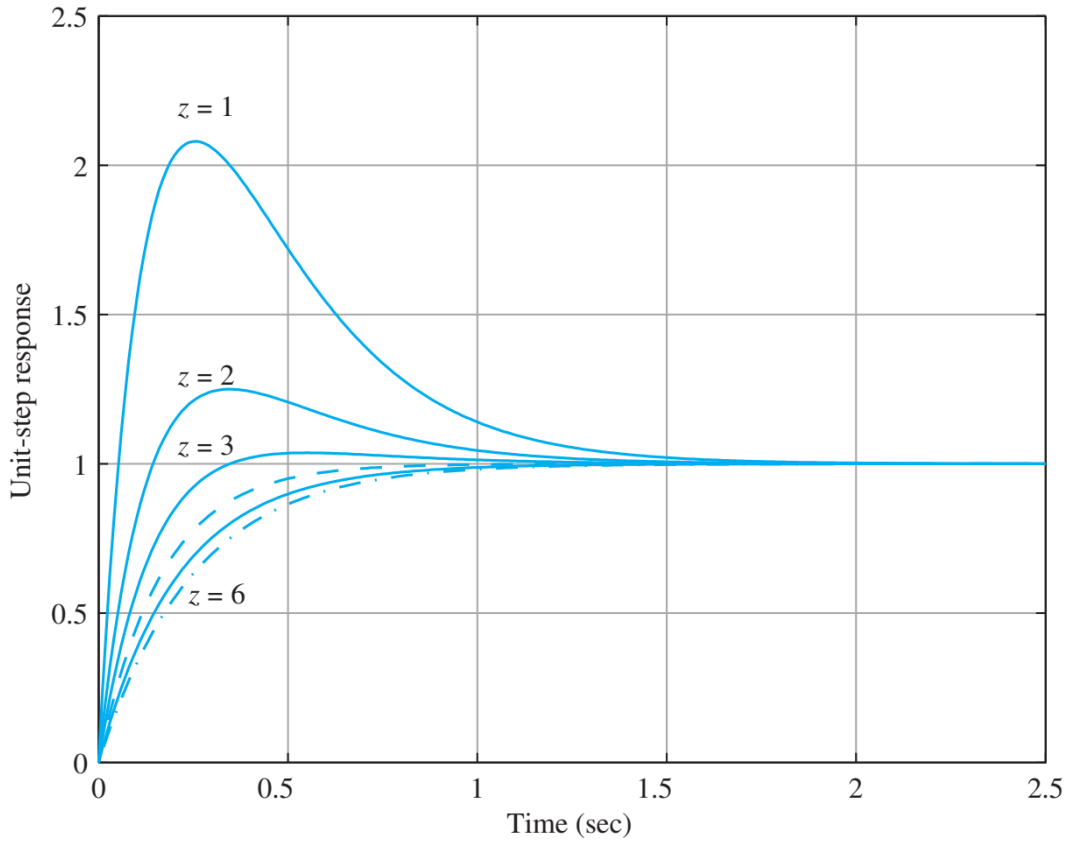
1. Effects of Zeros and Additional Poles

Example: Effect of the Proximity of the Zero to the Pole Locations on the Transient Response

- Additional overshoot for z = 1
- For z = 5 (between the two poles) there is no overshoot

$$H_1(s) = \frac{24}{z}\frac{s}{s(s+4)(s+6)} + \frac{24}{s(s+4)(s+6)}$$

$$y(t) = 1 + \left(\frac{12}{z} - 3\right)e^{-4t} + \left(2 - \frac{12}{z}\right)e^{-6t}$$



1. Effects of Zeros and Additional Poles

Example: Effect of the Proximity of the Complex Zeros to the Lightly Damped Poles (**Matlab**)

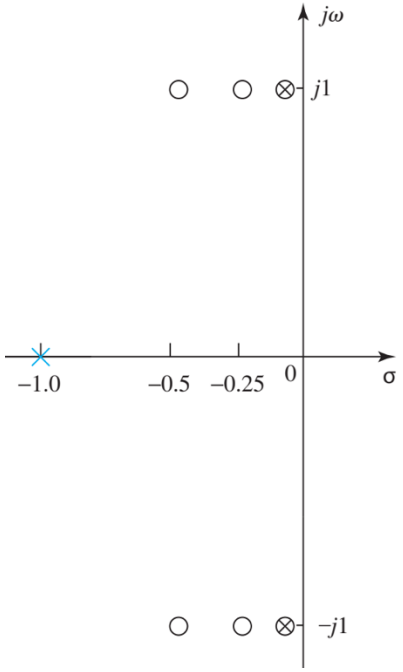
- 3rd order system
  - 1 real pole
  - 2 lightly damped complex poles
- 2 complex zeros
- Determine the effect of the complex zero locations ($s = -\alpha \pm j\beta$)
  - $(\alpha, \beta) = (0.1, 1.0)$, $(\alpha, \beta) = (0.25, 1.0)$, and $(\alpha, \beta) = (0.5, 1.0)$

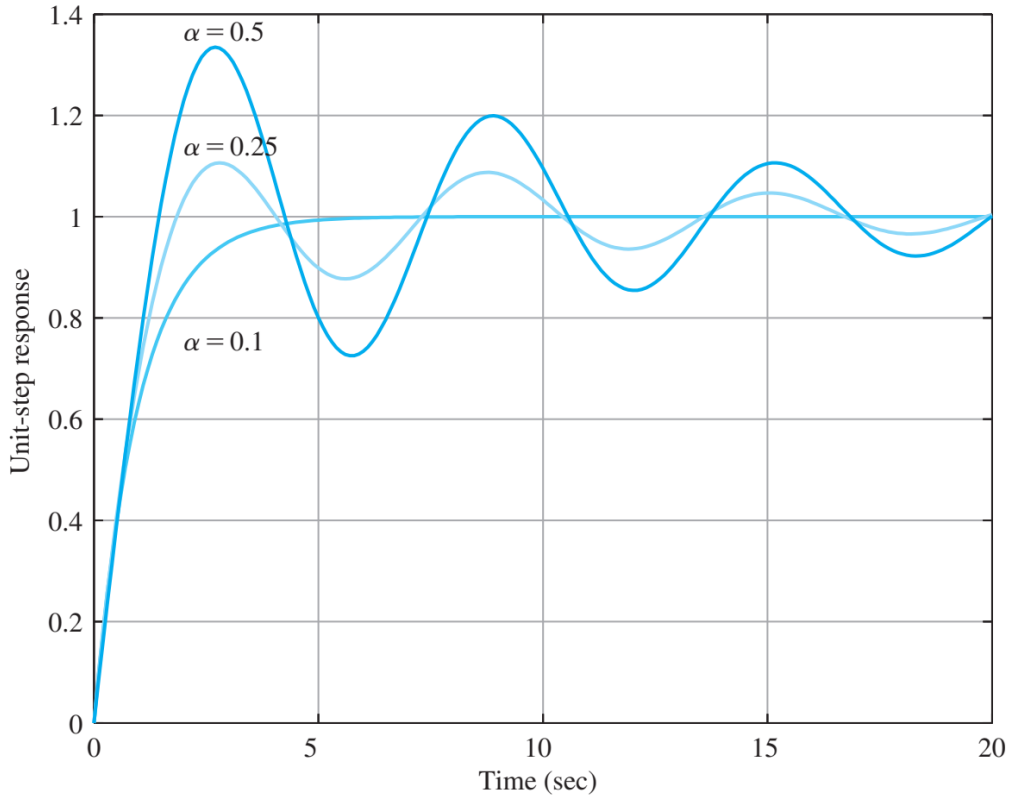
$$H(s) = \frac{(s+\alpha)^2 + \beta^2}{(s+1)\left[(s+0.1)^2 + 1\right]}$$

1. Effects of Zeros and Additional Poles

Example: Effect of the Proximity of the Complex Zeros to the Lightly Damped Poles (**Matlab**)

$$H(s) = \frac{(s+\alpha)^2 + \beta^2}{(s+1)\left[(s+0.1)^2 + 1\right]}$$

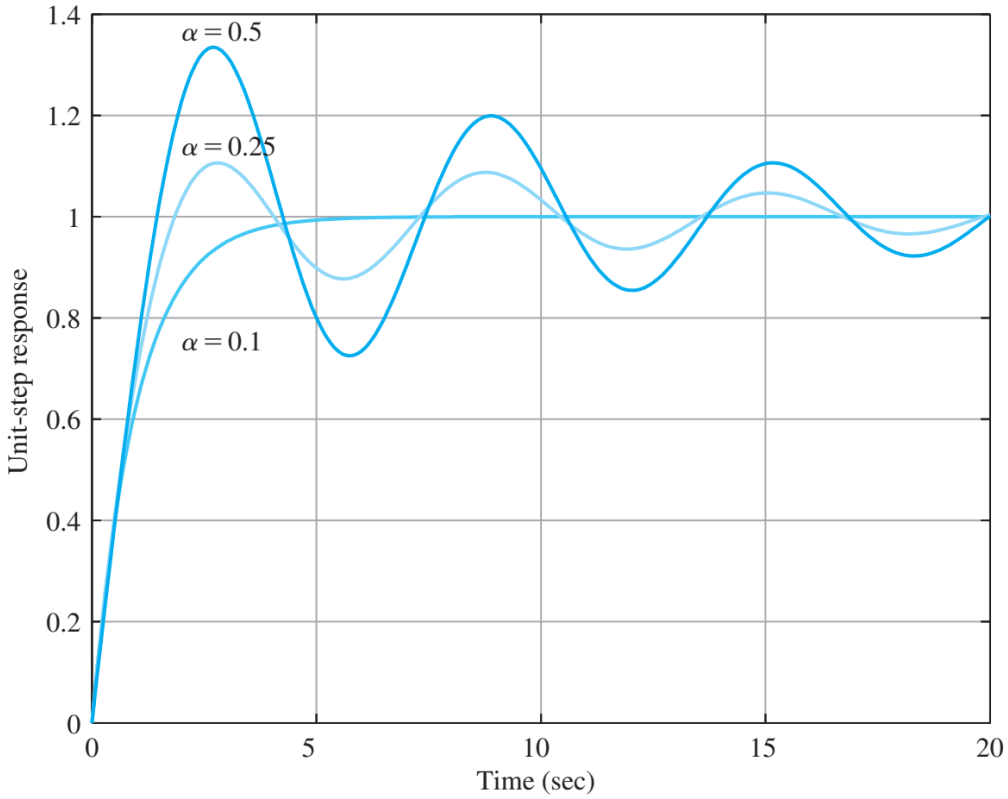




1. Effects of Zeros and Additional Poles

Example: Effect of the Proximity of the Complex Zeros to the Lightly Damped Poles (**Matlab**)

$$H(s) = \frac{(s+\alpha)^2 + \beta^2}{(s+1)\left[(s+0.1)^2 + 1\right]}$$



- In practice:
  - The **locations of the poles** are **not known precisely** => exact cancellation is not really possible
  - Placing the **complex zeros near the complex poles** may => **sufficient improvement** in the response

1. Effects of Zeros and Additional Poles

Example: Aircraft Response Using Matlab

Transfer function of of the Boeing 747 aircraft

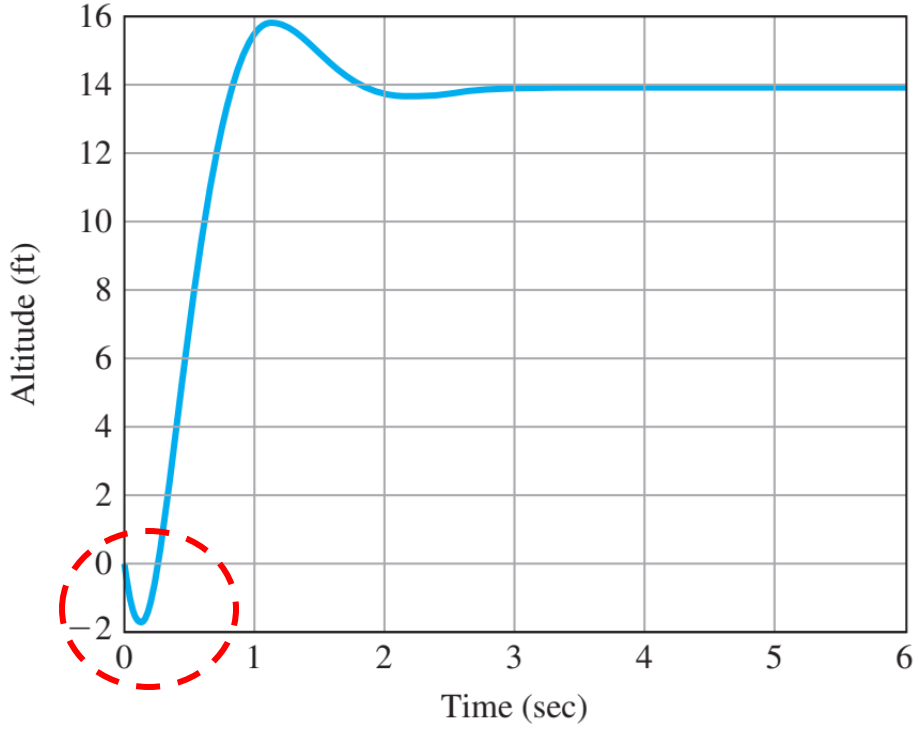
$$\frac{h(s)}{\delta_e(s)} = \frac{30(s-6)}{s(s^2+4s+13)}$$

1. Use Matlab to plot the altitude time history for a **impulsive** elevator input
2. Examine the accuracy of the approximations for tr, ts, and Mp

1. Effects of Zeros and Additional Poles

Example: Aircraft Response Using Matlab

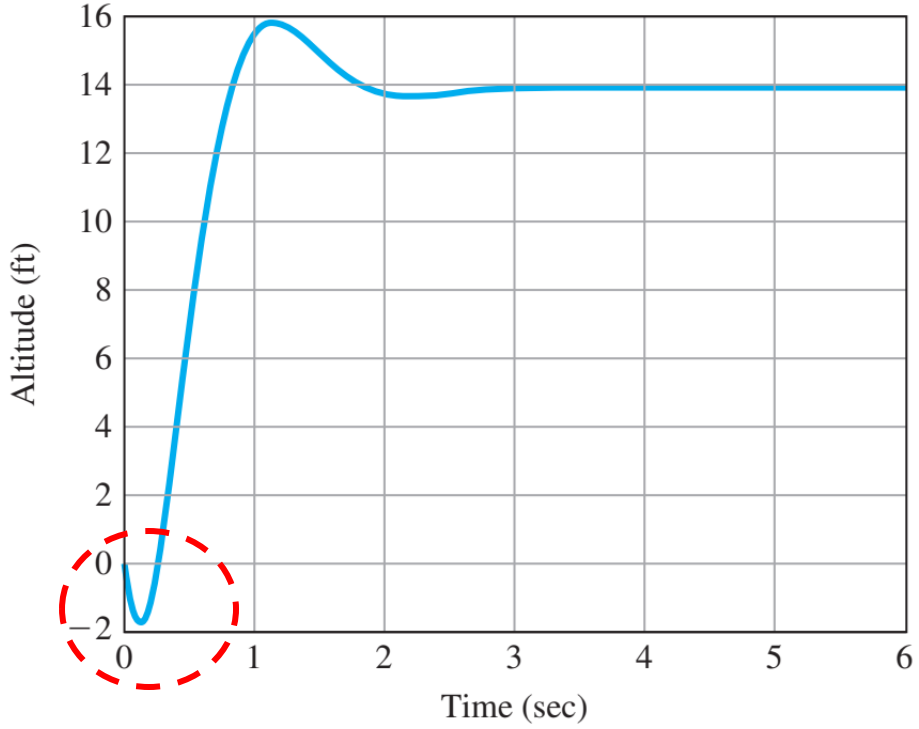
$$\frac{h(s)}{\delta_e(s)} = \frac{30(s-6)}{s(s^2+4s+13)}$$




```
u= -1; % u = delta e
s = tf('s');
% define system by its transfer function
sysG=u*30*(s-6)/(s^3+4*s^2+13*s);
y=impulse(sysG); % compute impulse response;
y = h
plot(y); % plot impulse response
```

1. Effects of Zeros and Additional Poles

Example: Aircraft Response Using Matlab



$$\frac{h(s)}{\delta_e(s)} = \frac{30(s-6)}{s(s^2+4s+13)}$$

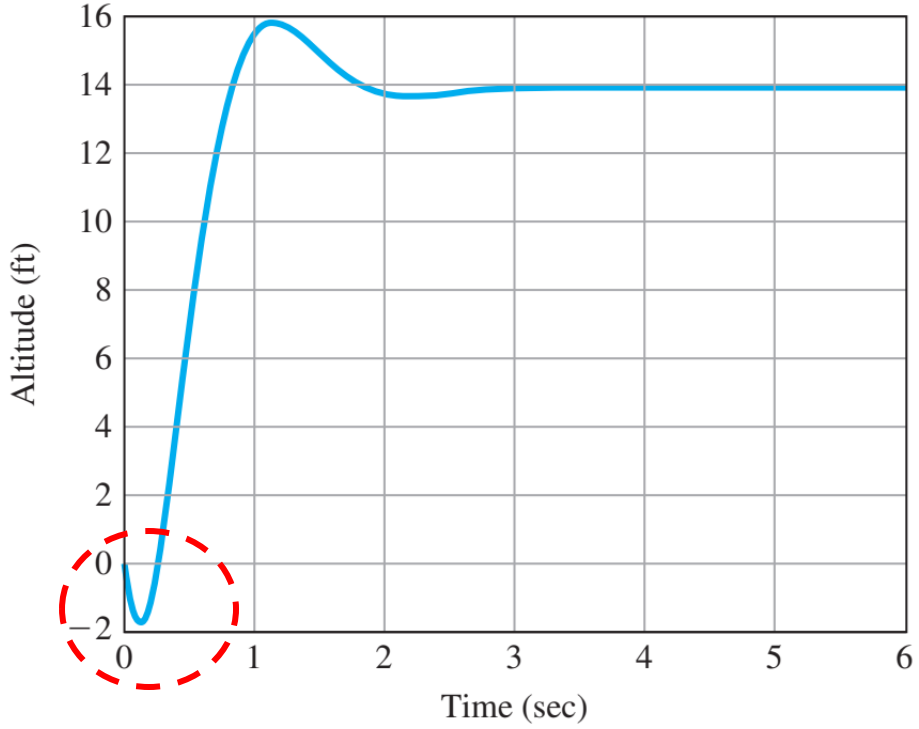
Steady-state value by using the Final-value theorem

$$h(\infty) = s\,\frac{30(s-6)(-1)}{s(s^2+4s+13)}\bigg|_{s=0} = \frac{30(-6)(-1)}{13} = +13.8.$$

- **Finite final value for an impulsive** due to the **integrator** (s in the denominator)
- The **initial drop** due to the **RHP** zero

1. Effects of Zeros and Additional Poles

Example: Aircraft Response Using Matlab

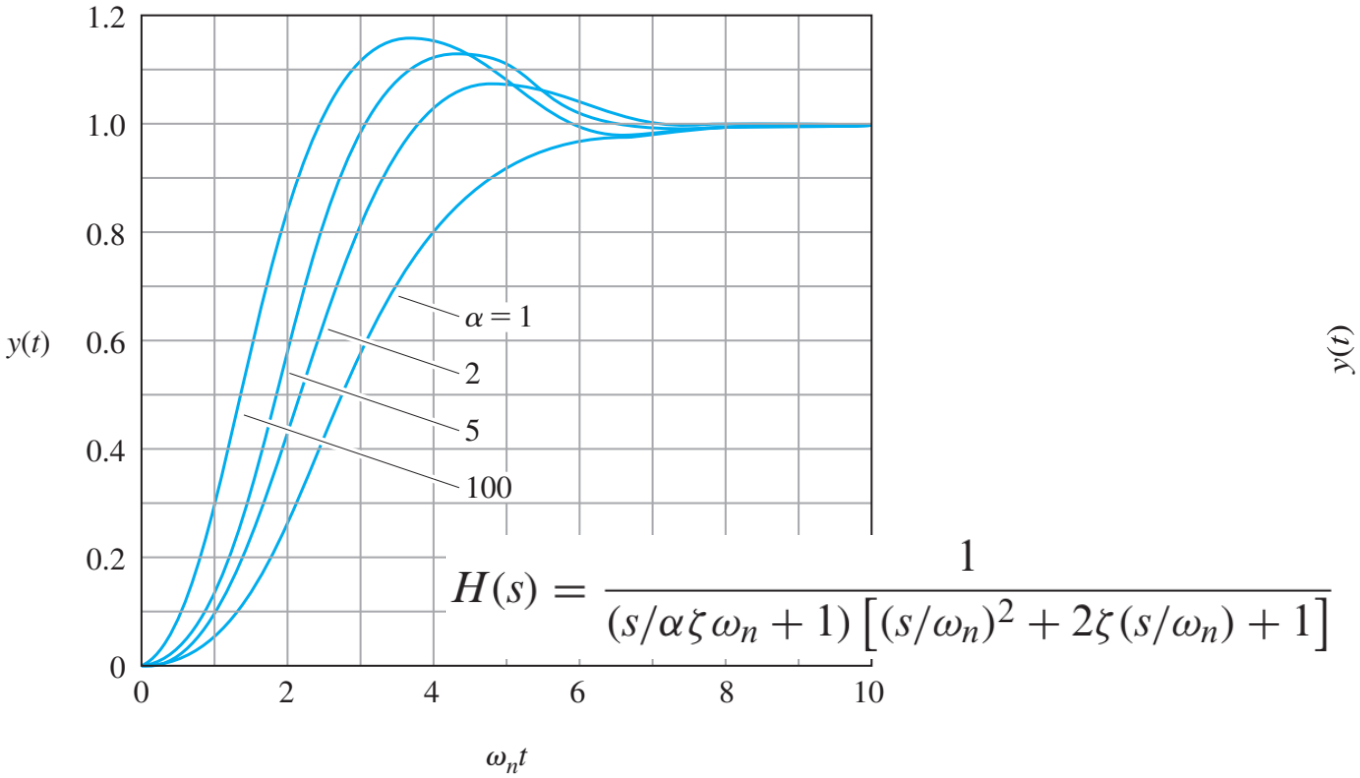


$$\frac{h(s)}{\delta_e(s)} = \frac{30(s-6)}{s(s^2+4s+13)}$$

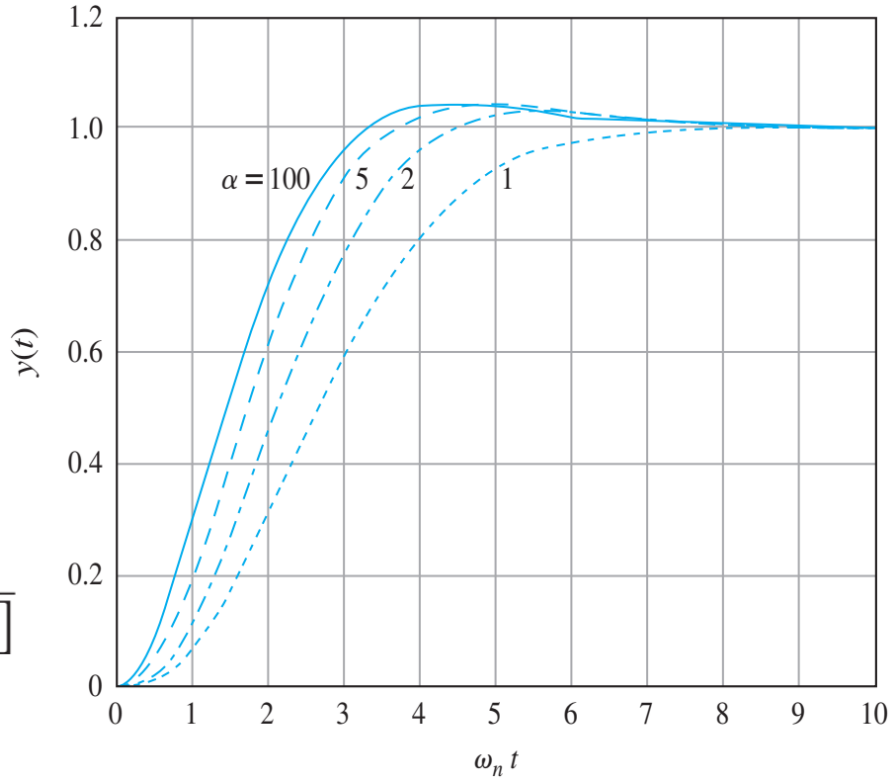
Estimated values	Matlab values
$t_r = \frac{1.8}{\omega_n} = \frac{1.8}{\sqrt{13}} = 0.5 \text{ sec.}$	tr ≈ 0.43 sec
M_p = 0.14 (from Figure)	Mp ≈ 0.14
$2\zeta\omega_n = 4 \Rightarrow \zeta = \frac{2}{\sqrt{13}} = 0.55$	
$t_s = \frac{4.6}{\sigma} = \frac{4.6}{2} = 2.3 \text{ sec.}$	ts ≈ 2.6 sec

1. Effects of Zeros and Additional Poles

Effects of an extra pole on the standard 2nd-order step response



$$H(s) = \frac{1}{(s/\alpha\zeta\omega_n + 1)\left[(s/\omega_n)^2 + 2\zeta(s/\omega_n) + 1\right]}$$

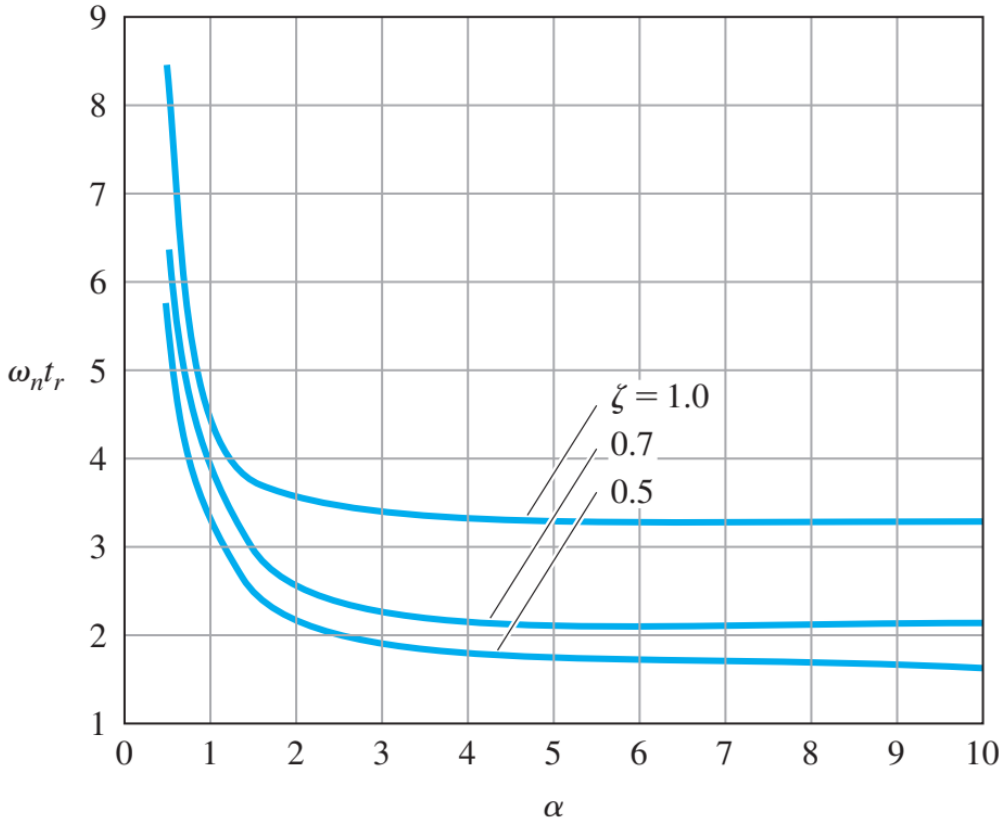


1. Effects of Zeros and Additional Poles

Effects of an extra pole on the standard 2nd-order step response

$$H(s) = \frac{1}{(s/\alpha\zeta\omega_n + 1)\left[(s/\omega_n)^2 + 2\zeta(s/\omega_n) + 1\right]}$$

The main effect is to increase the rise time



1. Effects of Zeros and Additional Poles

Summary:

- For a second-order system with no finite zeros tr, Mp, and ts can be approximated(see the equation)
- A zero in the LHP will increase the overshoot if the zero is within a factor of 4 of the real part of the complex poles
- A zero in the RHP will depress the overshoot and may cause the step response to start out in the wrong direction
- An additional pole in the LHP will increase the rise time significantly if the extra pole is within a factor of 4 of the real part of the complex poles

$$t_r \cong \frac{1.8}{\omega_n},$$

$$M_p \cong \begin{cases} 5\%, & \zeta = 0.7 \\ 16\%, & \zeta = 0.5 \\ 35\%, & \zeta = 0.3 \end{cases}$$

$$t_s \cong \frac{4.6}{\sigma}.$$

2. Stability

- Two basic ways to analyze stability
  - **Bounded Input–Bounded Output Stability**
  - **Routh's stability criterion**

2. Stability

- **Bounded Input–Bounded Output Stability**

- An **LTI** system is said to be **stable** if all **the roots** of the transfer function **denominator** polynomial have **negative real parts** (they are all in the left hand s-plane)
- Unstable otherwise
  - Any pole on the imaginary axis
  - Any pole in the RHP

2. Stability

Bounded Input–Bounded Output Stability

- The system with impulse response h(t) is **BIBO-stable** if and only if the following condition is true

$$\int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty.$$

2. Stability

Bounded Input–Bounded Output Stability

The output will be bounded if $\int_{-\infty}^{\infty} |h|\, d\tau$ is bounded

$$y(t) = \int_{-\infty}^{\infty} h(\tau)u(t-\tau)d\tau$$

$$|y| = \left| \int hu\, d\tau \right|$$

$$\leq \int |h||u|\, d\tau$$

$$\leq M \int_{-\infty}^{\infty} |h(\tau)|\, d\tau$$

The output will be bounded if $\int_{-\infty}^{\infty} |h|\, d\tau$ is bounded

Or, making:

$$u(t-\tau) = +1 \text{ if } h(\tau) > 0$$

$$u(t-\tau) = -1 \text{ if } h(\tau) < 0$$

$$y(t) = \int_{-\infty}^{\infty} |h(\tau)| d\tau$$

2. Stability

Bounded Input–Bounded Output Stability

Example: BIBO Stability for a Capacitor with

- Input: current u(t)
- Output: voltage y(t)



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$$h(t) = 1(t)$$

$$\int_{-\infty}^{\infty} |h(\tau)| d\tau = \int_{0}^{\infty} d\tau$$

y(t) not bounded => not stable

2. Stability

- Consider T(s) where pi are all different and before any cancellation
- If any poles are repeated => change Ki by t, but the conclusion is the same
- If a **zero** were to **cancel** a **pole** in the **RHP** the corresponding Ki would equal zero, but **unstable transient would appear in some internal** variable
- If the system has **nonrepeated** jω axis poles, then **neutrally stable**
- If the system has **repeated** jω axis poles, then **unstable**

$$T(s) = \frac{Y(s)}{R(s)} = \frac{b_0 s^m + b_1 s^{m-1} + \cdots + b_m}{s^n + a_1 s^{n-1} + \cdots + a_n}$$

$$= \frac{K \prod_{i=1}^{m} (s - z_i)}{\prod_{i=1}^{n} (s - p_i)}, \quad m \leq n.$$

$$s^n + a_1 s^{n-1} + a_2 s^{n-2} + \cdots + a_n = 0.$$

$$y(t) = \sum_{i=1}^{n} K_i e^{p_i t}$$

y(t) -> if

$$e^{p_i t} \to 0 \text{ for all } p_i$$

2. Stability

- Does not require the calculation of the roots of the characteristic equations
- It is useful in determining the **range of parameters** for which a **feedback system** remains **stable**
- **Necessary (but not sufficient) condition for stability**
  - **{ai} be positive**
- If any of **{ai}** are **0** (missing) or **negative** => there are poles located outside the LHP

Characteristic equation of H(s)

$$a(s) = s^n + a_1 s^{n-1} + a_2 s^{n-2} + \cdots + a_{n-1} s + a_n.$$

2. Stability

Routh's Stability Criterion

A system is stable if and only if all the elements in the first column of the Routh array are positive

Row	n	s^n:	1	a_2	a_4	$\cdots$
Row	$n-1$	s^{n-1}:	a_1	a_3	a_5	$\cdots$
Row	$n-2$	s^{n-2}:	b_1	b_2	b_3	$\cdots$
Row	$n-3$	s^{n-3}:	c_1	c_2	c_3	$\cdots$
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	
Row	2	s^2:	$*$	$*$		
Row	1	s^1:	$*$			
Row	0	s^0:	$*$			

$$b_1 = -\frac{\det\begin{bmatrix} 1 & a_2 \\ a_1 & a_3 \end{bmatrix}}{a_1} = \frac{a_1 a_2 - a_3}{a_1},$$

$$b_2 = -\frac{\det\begin{bmatrix} 1 & a_4 \\ a_1 & a_5 \end{bmatrix}}{a_1} = \frac{a_1 a_4 - a_5}{a_1},$$

$$b_3 = -\frac{\det\begin{bmatrix} 1 & a_6 \\ a_1 & a_7 \end{bmatrix}}{a_1} = \frac{a_1 a_6 - a_7}{a_1},$$

$$c_1 = -\frac{\det\begin{bmatrix} a_1 & a_3 \\ b_1 & b_2 \end{bmatrix}}{b_1} = \frac{b_1 a_3 - a_1 b_2}{b_1}$$

$$c_2 = -\frac{\det\begin{bmatrix} a_1 & a_5 \\ b_1 & b_3 \end{bmatrix}}{b_1} = \frac{b_1 a_5 - a_1 b_3}{b_1}$$

$$c_3 = -\frac{\det\begin{bmatrix} a_1 & a_7 \\ b_1 & b_4 \end{bmatrix}}{b_1} = \frac{b_1 a_7 - a_1 b_4}{b_1}$$

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	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	
Row	2	s^2:	$*$	$*$		
Row	1	s^1:	$*$			
Row	0	s^0:	$*$			

- If the elements of the first column are not all positive => the number of **roots** in the **RHP** equals the **number of sign changes** in the 1st columna
- If only the **first element** in **one** of the **rows** of the Routh array is **zero** or an **entire row** of the Routh array is **zero =>** the method has some modifications **(special case)**

2. Stability

Routh's Stability Criterion

Example: evaluate the stability of a transfer function with the following characteristic equation

$$a(s) = s^6 + s^5 + s^4 + 2s^3 + 5s^2 + 2s + 3$$

2. Stability

Routh's Stability Criterion

Example: evaluate the stability of a transfer function with the following characteristic equation

$$a(s) = s^6 + s^5 + s^4 + 2s^3 + 5s^2 + 2s + 3$$

s^6:	1	1	5	3
s^5:	1	2	2	0
s^4:	$-1 = \frac{1 \cdot 1 - 1 \cdot 2}{1}$	$3 = \frac{1 \cdot 5 - 1 \cdot 2}{1}$	$3 = \frac{1 \cdot 3 - 1 \cdot 0}{1}$	
s^3:	$5 = \frac{-1 \cdot 2 - 1 \cdot 3}{-1}$	$5 = \frac{-1 \cdot 2 - 1 \cdot 3}{-1}$	0	
s^2:	$4 = \frac{5 \cdot 3 + 1 \cdot 5}{5}$	$3 = \frac{5 \cdot 3 + 1 \cdot 0}{5}$		
s:	$\frac{5}{4} = \frac{4 \cdot 5 - 5 \cdot 3}{4}$	0		
s^0:	$3 = \frac{\frac{5}{4} \cdot 3 - 4.0}{\frac{5}{4}}$			

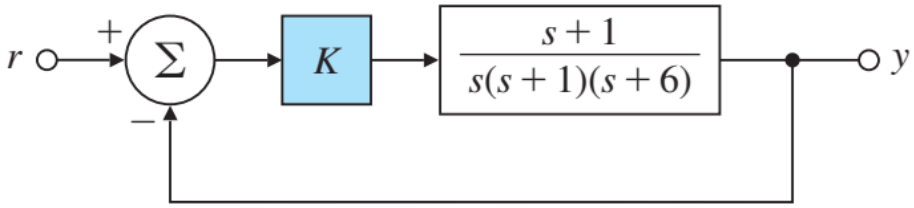
2 poles in the RHP because there are two sign changes

2. Stability

Routh's Stability Criterion

Example: Stability versus Parameter Range

- Determine the range of K over which the system is stable

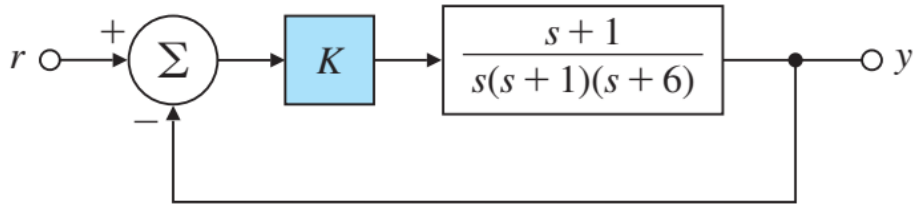


2. Stability

Routh's Stability Criterion

Example: Stability versus Parameter Range

- Determine the range of K over which the system is stable



Characteristic equation

$$1 + K\frac{s+1}{s(s-1)(s+6)} = 0, \qquad s^3 + 5s^2 + (K-6)s + K = 0.$$

Routh array:

$$\begin{array}{lcc} s^3: & 1 & K-6 \\ s^2: & 5 & K \\ s: & (4K-30)/5 & \\ s^0: & K. & \end{array}$$

The system is stable if:

$$\frac{4K-30}{5} > 0 \quad \text{and} \quad K > 0,$$

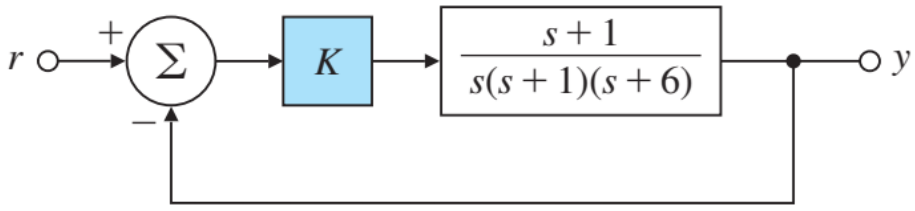
$$K > 7.5 \quad \text{and} \quad K > 0$$

2. Stability

Routh's Stability Criterion

Example: Stability versus Parameter Range

- Determine the range of K over which the system is stable



$$T(s) = \frac{Y(s)}{R(s)} = \frac{K(s+1)}{s^3 + 5s^2 + (K-6)s + K}$$

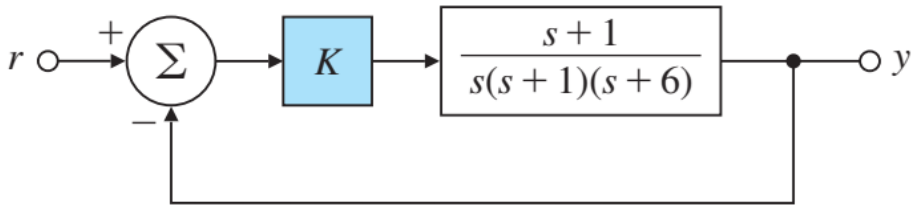
```
K_vec = [7.5 13 25];
s=tf('s'); % define the Laplace variable
for i = 1:length(K_vec)
K = K_vec(i);
denT = [1 5 K-6 K];
aux = roots(denT);
disp(aux);
sysT=K*(s+1)/(s^3+5*s^2+(K-6)*s+K); % define transfer function
step(sysT,15); % compute the step response
hold on;
end
hold off;
```

2. Stability

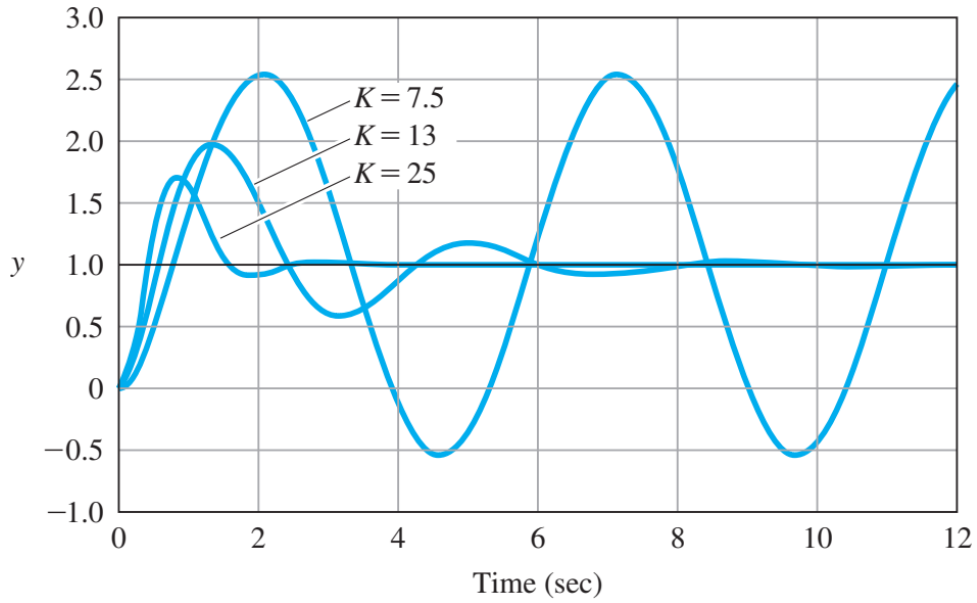
Routh's Stability Criterion

Example: Stability versus Parameter Range

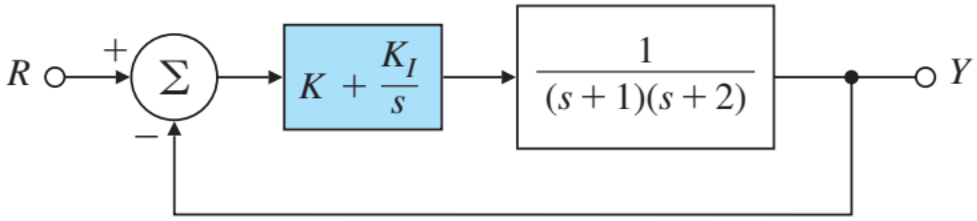
- Determine the range of K over which the system is stable



K = 7.5 => poles in jw
K = 13 => poles −4.06 and −0.47 ± 1.7j
K = 25 => poles −1.90 and −1.54 ± 3.27j



2. Stability



Routh's Stability Criterion

Example: Stability versus 2 Parameter Ranges

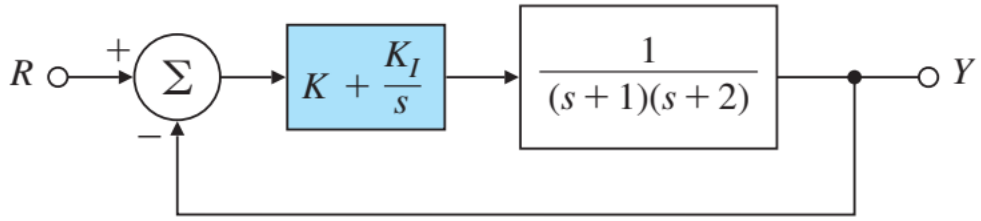
- Determine the ranges of K and K_I over which the system is stable

2. Stability

Routh's Stability Criterion

Example: Stability versus 2 Parameter Ranges

- Determine the ranges of K and K_I over which the system is stable



Characteristic equation

$$1 + \left(K + \frac{K_I}{s}\right)\frac{1}{(s+1)(s+2)} = 0 \qquad s^3 + 3s^2 + (2+K)s + K_I = 0$$

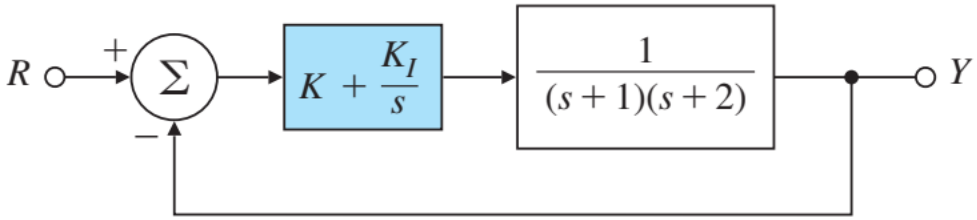
Routh array:

s^3 :	1	$2+K$
s^2 :	3	K_I
s :	$(6+3K-K_I)/3$	
s^0 :	K_I.	

The system is stable if:

$$K_I > 0 \quad \text{and} \quad K > \frac{1}{3}K_I - 2$$

2. Stability



Routh's Stability Criterion

Example: Stability versus 2 Parameter Ranges

- Determine the ranges of K and K_I over which the system is stable

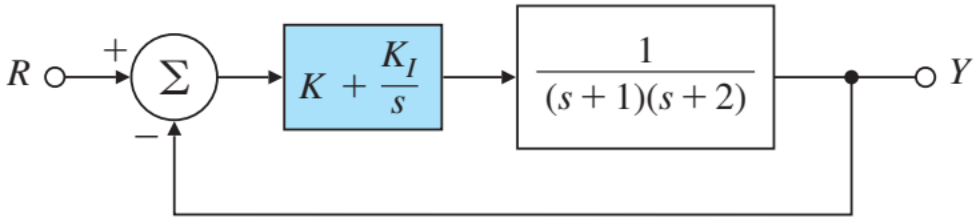
```
figure;
fh=@(KI,K) 6+3*K-KI;
ezplot(fh)
hold on;
f=@(KI,K) KI;
ezplot(f);
hold off;
```

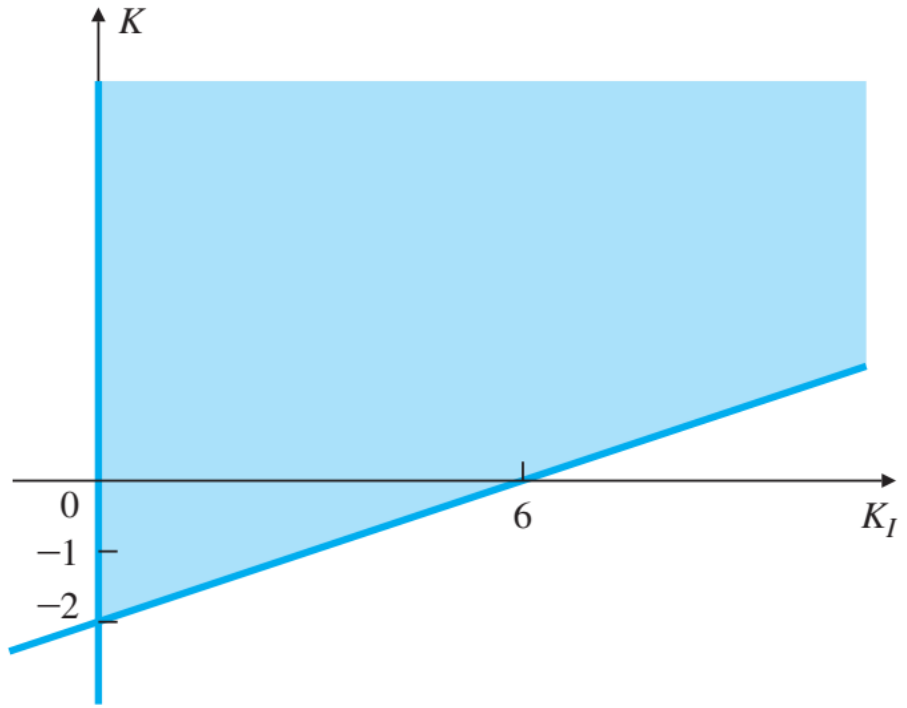
2. Stability

Routh's Stability Criterion

Example: Stability versus 2 Parameter Ranges

- Determine the ranges of K and K_I over which the system is stable



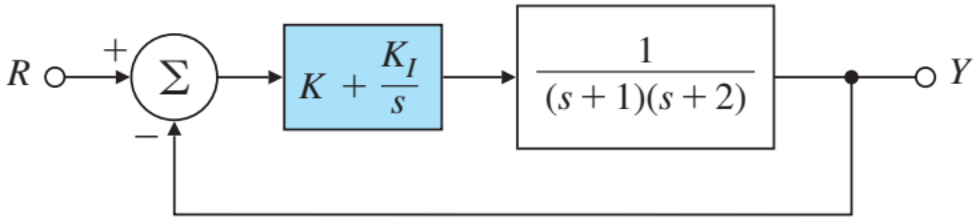


2. Stability

Routh's Stability Criterion

Example: Stability versus 2 Parameter Ranges

- Determine the ranges of K and K_I over which the system is stable



$$T(s) = \frac{Y(s)}{R(s)} = \frac{Ks + K_I}{s^3 + 3s^2 + (2+K)s + K_I}$$

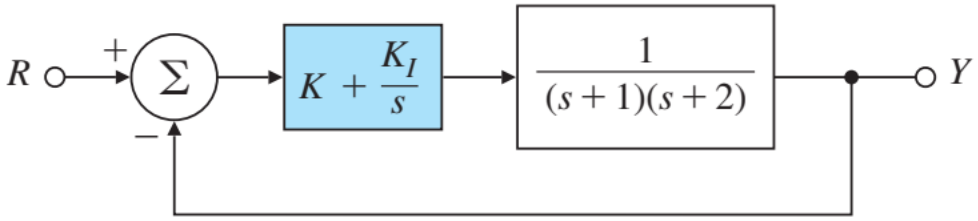
K	K_I	Poles	Zero
1	0	0 and −1.5 ± 0.86j	0
1	1	3 poles at −1	-1
10	5	−0.46 and −1.26 ± 3.3j	-0.5

2. Stability

Routh's Stability Criterion

Example: Stability versus 2 Parameter Ranges

- Determine the ranges of K and K_I over which the system is stable



$$T(s) = \frac{Y(s)}{R(s)} = \frac{Ks + K_I}{s^3 + 3s^2 + (2+K)s + K_I}$$

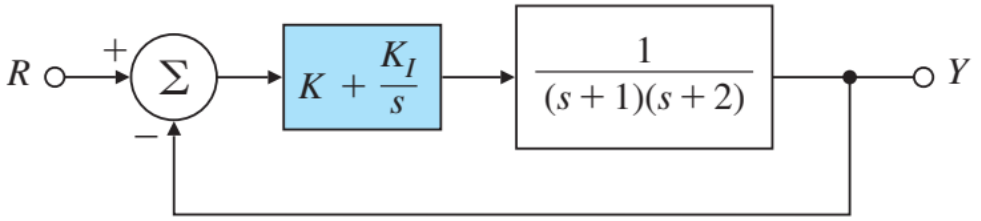
```
figure;
K_vec = [1 1 10];
KI_vec = [0 1 5];
for i = 1:length(KI_vec)
K = K_vec(i);
KI = KI_vec(i);
denT = [1 3 2+K KI]; % form denominator
numT = [K KI]; % form numerator
sysT=tf(numT,denT); % define system by its numerator and
denominator
step(sysT,10) % compute step response
hold on;
end
hold off;
```

2. Stability

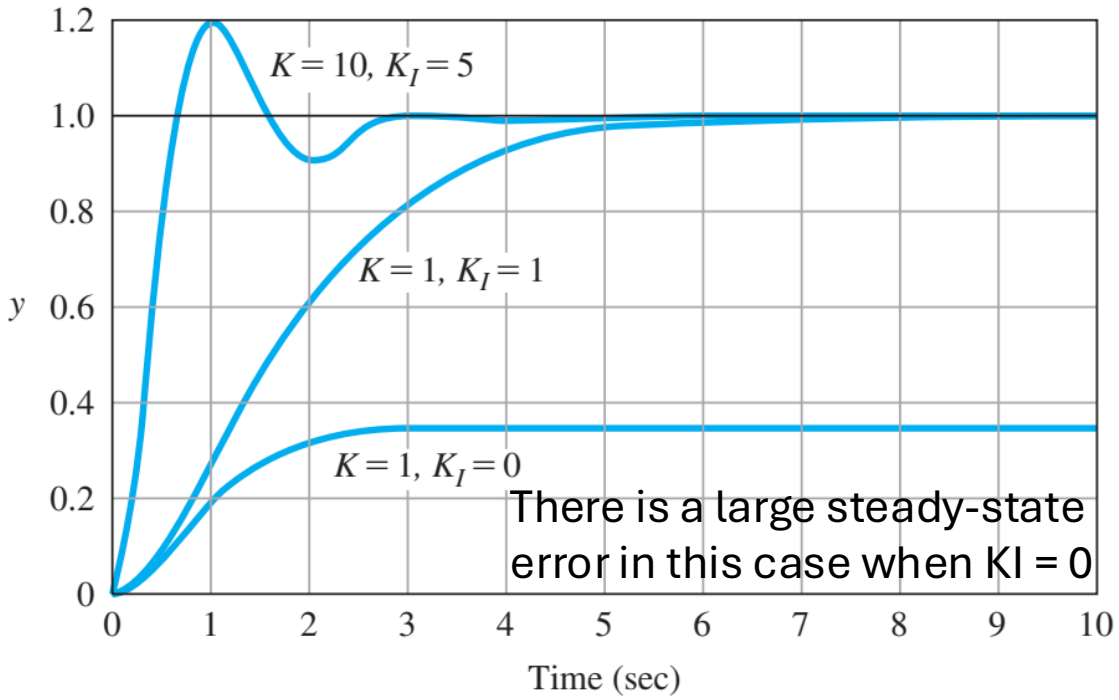
Routh's Stability Criterion

Example: Stability versus 2 Parameter Ranges

- Determine the ranges of K and K_I over which the system is stable



$$T(s) = \frac{Y(s)}{R(s)} = \frac{Ks + K_I}{s^3 + 3s^2 + (2+K)s + K_I}$$



There is a large steady-state error in this case when KI = 0